

SULAMAN KHAN

AI Engineer | LLM Applications, RAG & Agentic Systems

Remote (US / Gulf time zones) • Lahore, Pakistan • +92-316-7852626 • se.sulaman.khan@gmail.com

github.com/sulaman0 • linkedin.com/in/sulaman-khan

PROFESSIONAL SUMMARY

AI Engineer with 8 years of production software experience and the last 2 years focused on integrating AI into real systems. Hands-on experience delivering Retrieval-Augmented Generation (RAG) pipelines, Large Language Model (LLM) agents with tool calling, Model Context Protocol (MCP) integrations, structured-output extraction, and computer-vision pipelines (object detection, OCR, face recognition) — engineered with the reliability, multi-tenant security, evaluation, and cost discipline that production demands. Backed by a track record in fintech platforms processing \$1M+ in transactions and systems serving 100,000+ daily users. Fluent in Python, Go, and Node.js. I bring rare depth: proven production engineering judgment applied to the AI layer, not AI demos.

CORE SKILLS

AI / LLM: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LLM Agents, Agentic Systems, Tool Calling, Prompt Engineering, Structured Outputs, Embeddings, Semantic Search, RAG Evaluation, Model Context Protocol (MCP), Multi-Tenant AI Security

AI Frameworks & Tools: LangChain, LangGraph, Anthropic Claude API, OpenAI API, Ollama, ChromaDB, pgvector, Vector Databases, FastAPI, Server-Sent Events (SSE)

Computer Vision: YOLO (Object Detection), Optical Character Recognition (OCR), ArcFace (Face Recognition), OpenCV

Backend: Python, Go (Golang), Node.js, Rust, TypeScript, gRPC, Microservices, REST APIs, Concurrency

Data & Cloud: PostgreSQL, pgvector, Amazon Web Services (AWS: Lambda, API Gateway), Docker, Kubernetes

Practices: Observability, Guardrails, Cost Optimization, CI/CD, System Design, Data Security

PROFESSIONAL EXPERIENCE

AI Engineer 2024 – Present

Vaival Technologies LLC — Remote

- **RAG pipelines:** Built end-to-end Retrieval-Augmented Generation (RAG) systems — document chunking, embedding generation, and vector storage/retrieval (ChromaDB, pgvector) — that ground LLM responses in source data and eliminate hallucination, validated with an evaluation harness measuring retrieval hit-rate and answer faithfulness.
- **Multi-tenant AI security:** Implemented metadata-filtered retrieval enforcing strict tenant isolation, verified with security tests proving zero cross-tenant data leakage — directly applicable to regulated fintech and banking environments.
- **LLM agents & MCP:** Developed agentic systems using the ReAct reasoning loop (tool calling with step limits and input validation) and built Model Context Protocol (MCP) servers and clients that expose internal tools to any MCP-compatible AI client.
- **Production deployment:** Shipped RAG as an async FastAPI service with token streaming (Server-Sent Events), per-request cost and latency observability, provider-failure handling, and tenant isolation enforced at the API layer.
- **Reliability engineering:** Applied backend rigor to AI — retry logic with exponential backoff, structured-output schema enforcement with defensive parsing, graceful degradation on malformed model output, and multi-provider abstraction across Anthropic Claude and OpenAI.

Blockchain & Backend Engineer Aug 2021 – 2024

Vaival Technologies LLC

- **Backend at scale:** Built Go and Node.js backend services for a Layer 1 blockchain platform — explorer, decentralized exchange (DEX), NFT marketplace, bridge, and swap modules — managing real-time on-chain data via GraphQL.
- **Secure transactions:** Implemented transaction signing, DeFi protocol integration, and secure smart-contract interaction — a security-first discipline that transfers directly to production AI systems.
- **High performance:** Boosted throughput using Go concurrency (goroutines, channels) and gRPC microservices for reliable distributed communication.

Full Stack Engineer (Web & Mobile) Jun 2018 – Aug 2021

Vaival Technologies LLC / B4USoft Pvt. Ltd.

- **Fintech:** Built secure fintech features (transactions, authentication, data security); launched application processed \$1M+ in transactions in its first quarter.
- **High-traffic systems:** Designed and optimized a CMS supporting 100,000+ daily active users; built React Native e-commerce apps with Go-powered APIs, payments (Stripe, PayPal), and inventory.

- **Cloud & deployment:** Serverless architecture on AWS Lambda; Docker-based deployments; monitoring with Datadog — reduced operational costs 40% while improving scalability.

AI PROJECTS

- **Multi-Tenant RAG Service** — Production RAG API (Python, FastAPI, ChromaDB, Ollama) with token streaming via Server-Sent Events, tenant-isolated retrieval enforced per request, cost/latency observability middleware, and provider-failure handling. Demonstrates RAG delivered as a secure, scalable service.
- **MCP Server & Client** — Built both halves of a Model Context Protocol integration: a server exposing tools and resources, and a client that discovers and calls them at runtime, wired into a local-model agent loop. Shows the modern standard for connecting AI to internal systems.
- **Agentic System (ReAct)** — LLM agent with a think-act-observe tool-calling loop, step-limit safety caps, and schema validation at the tool boundary; implemented in raw orchestration and rebuilt in LangGraph to demonstrate both fundamentals and framework fluency.
- **Support Ticket Triage Engine** — LLM system that classifies priority, extracts structured fields, and drafts replies from raw tickets, featuring graceful degradation and clean engine/interface separation for production use.
- **Retail Shelf Detection & Recognition** — Computer-vision pipeline integrating YOLO for shelf and product detection, Optical Character Recognition (OCR) for on-shelf text extraction, and ArcFace for face recognition — combining multiple deep-learning models into one working system. (Python)
- **ZATCA Phase 2 E-Invoicing** — Saudi Arabia e-invoicing compliance implementation (Fatoora, XML generation, certificate signing) — direct experience with Kingdom of Saudi Arabia (KSA) regulatory requirements.

KEY ACHIEVEMENTS

- Fintech application processed **\$1M+ in transactions** within the first quarter post-launch.
- Scaled a CMS platform to **100,000+ daily active users**.
- Achieved **40% reduction in operational costs** via serverless architecture while improving scalability.

EDUCATION & CERTIFICATIONS

B.S. Information Technology 2014 – 2018

University of the Punjab, Lahore, Pakistan

- Applied Concurrency in Go — goroutines, channels, concurrent systems design (2023)